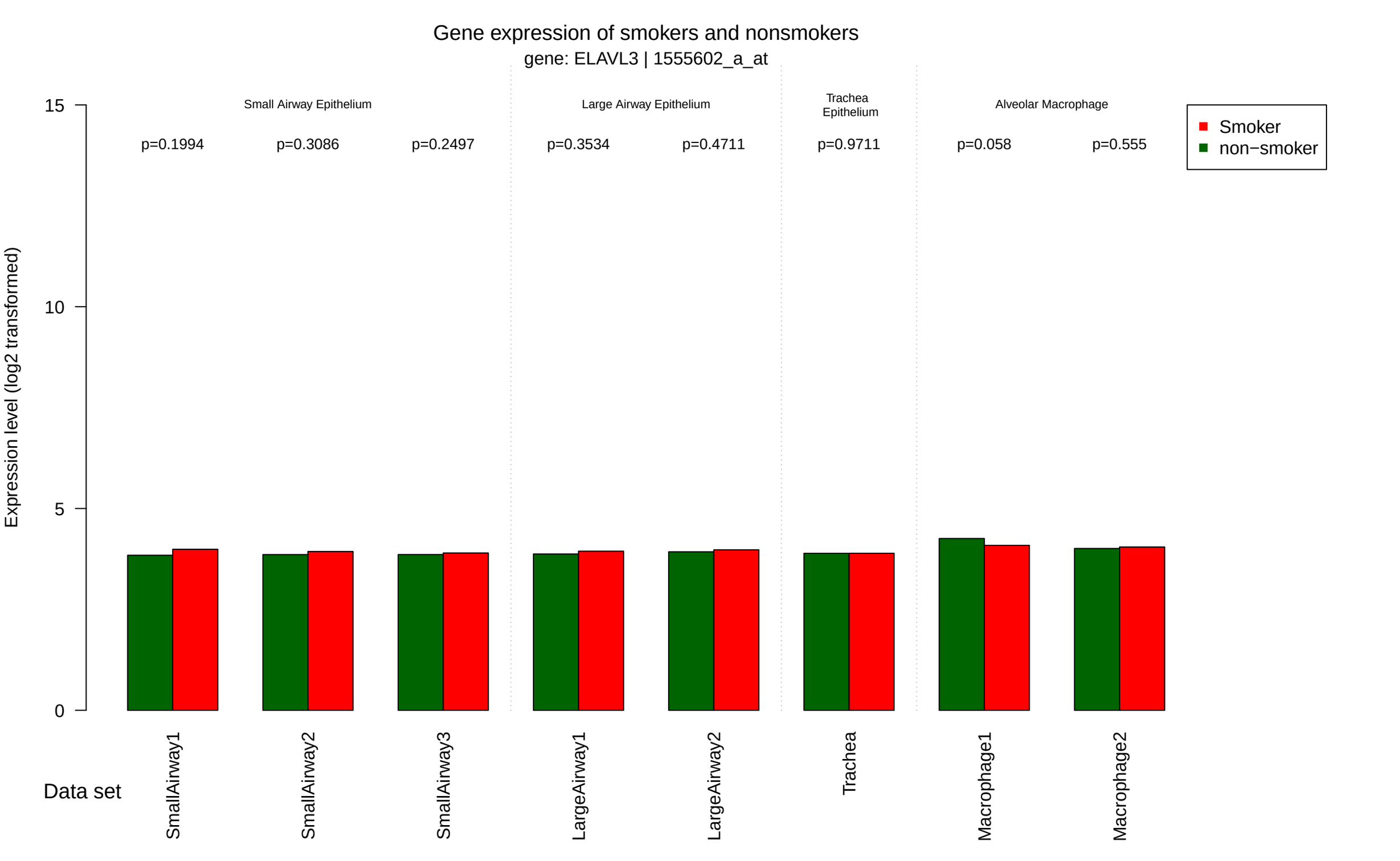
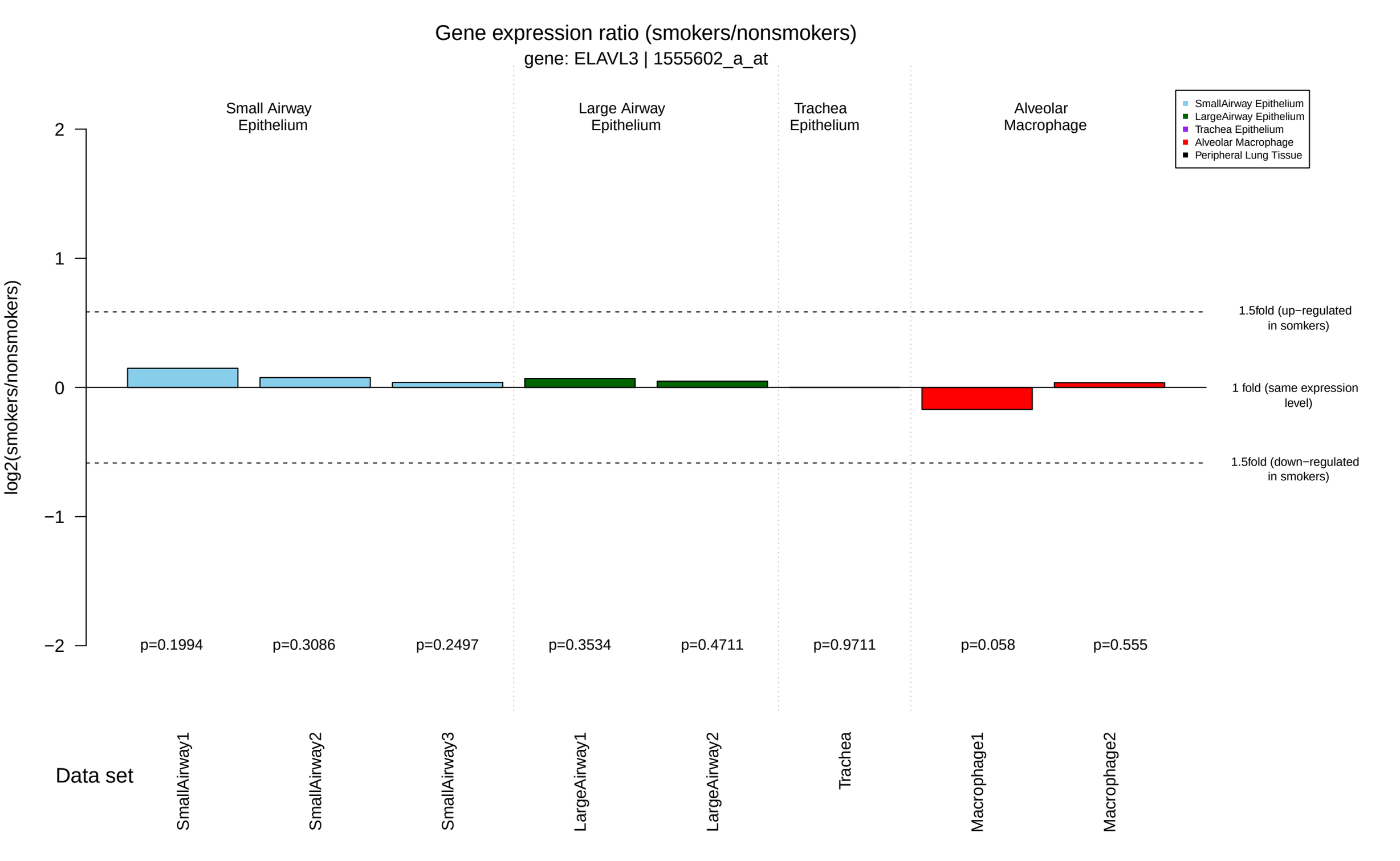
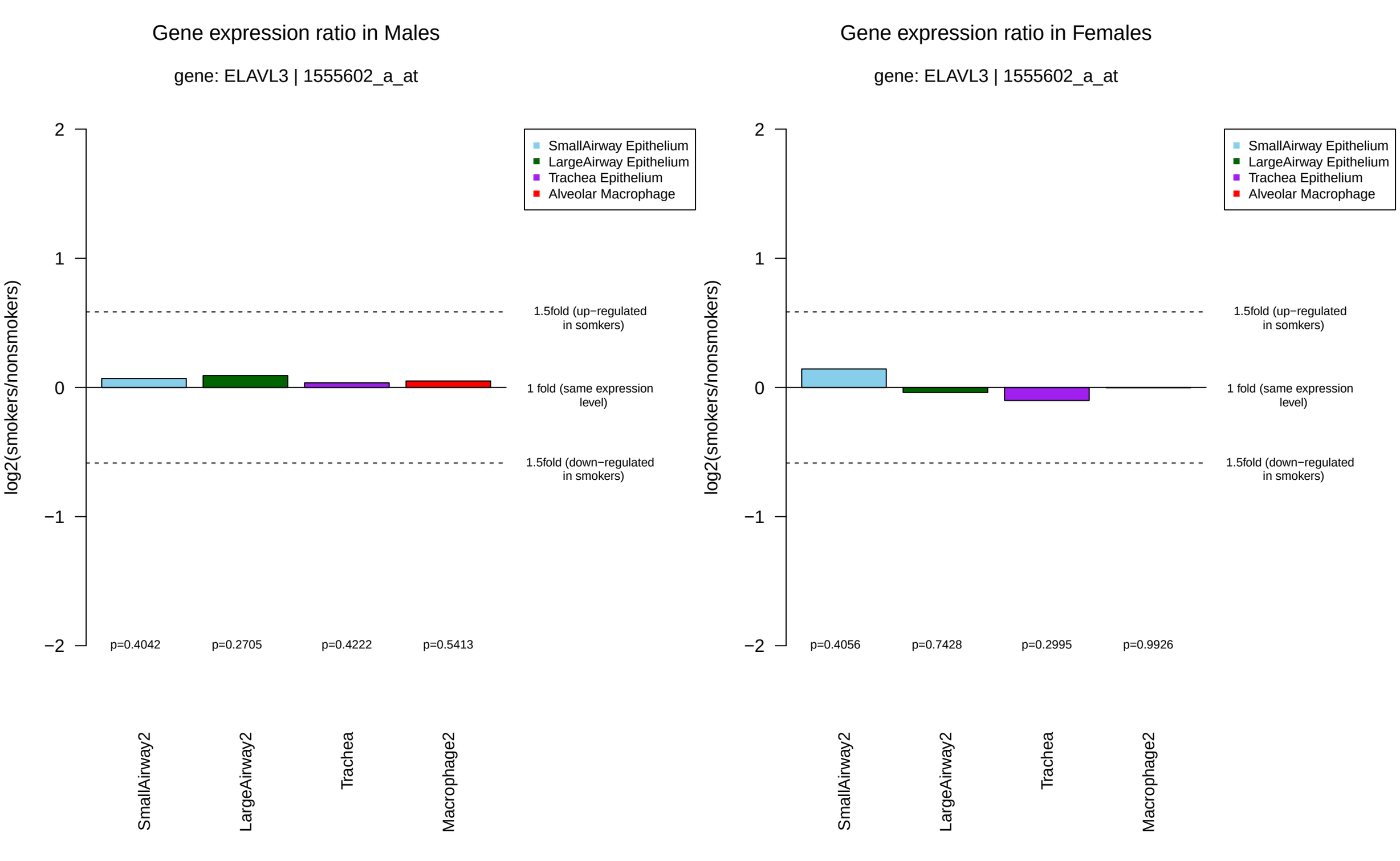
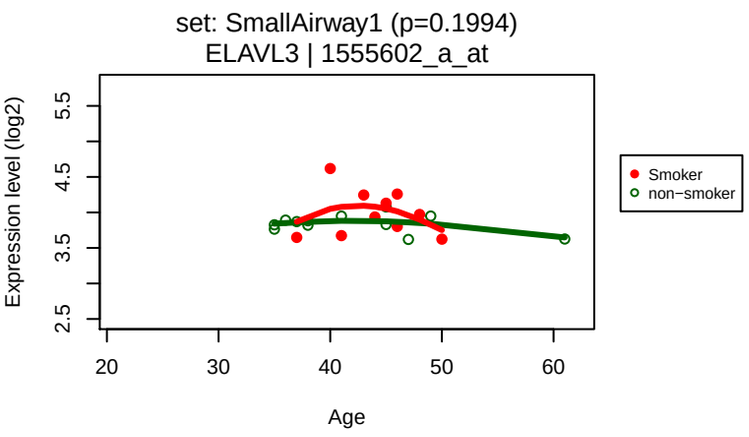
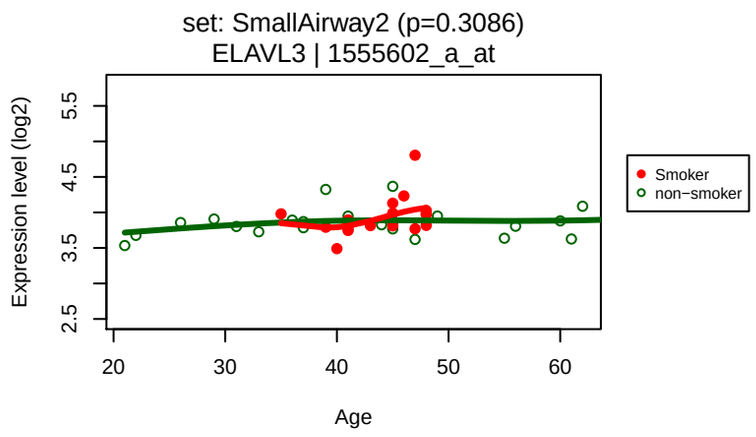
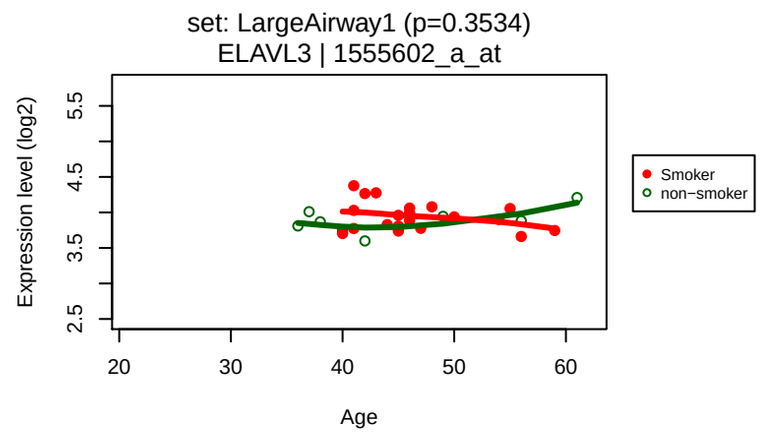

Gene expression of smokers and nonsmokers
gene: ELAVL3 | 1555602_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1994
p=0.3086
p=0.2497
p=0.3534
p=0.4711
p=0.9711
p=0.058
p=0.555
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

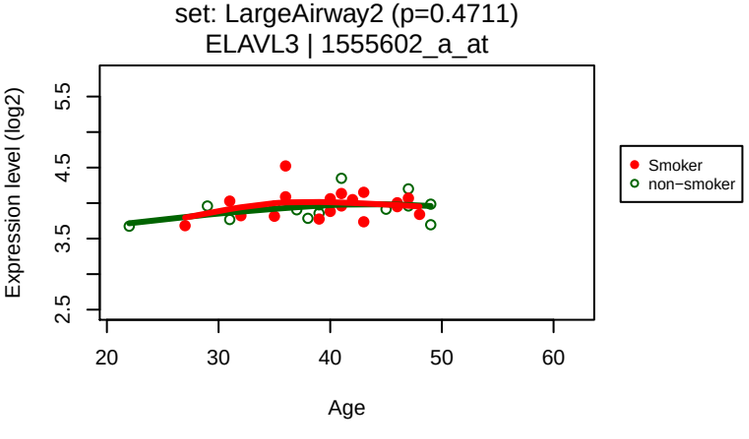

Gene expression ratio (smokers/nonsmokers)
gene: ELAVL3 | 1555602_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1994
p=0.3086
p=0.2497
p=0.3534
p=0.4711
p=0.9711
p=0.058
p=0.555
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

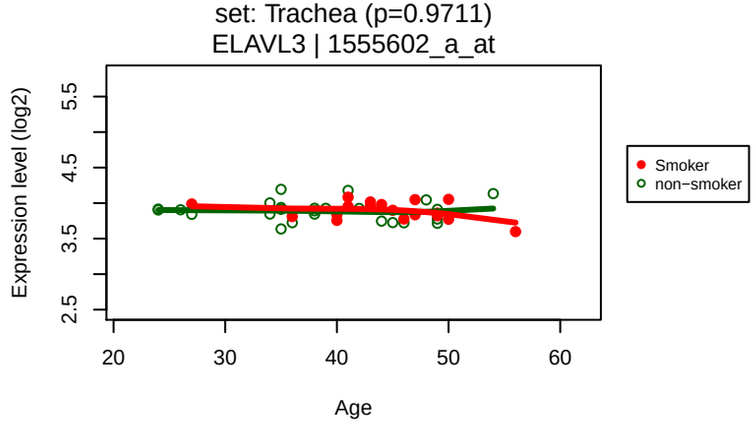

Gene expression ratio in Males
gene: ELAVL3 | 1555602_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4042
p=0.2705
p=0.4222
p=0.5413
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ELAVL3 | 1555602_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4056
p=0.7428
p=0.2995
p=0.9926
SmallAirway2
LargeAirway2
Trachea
Macrophage2

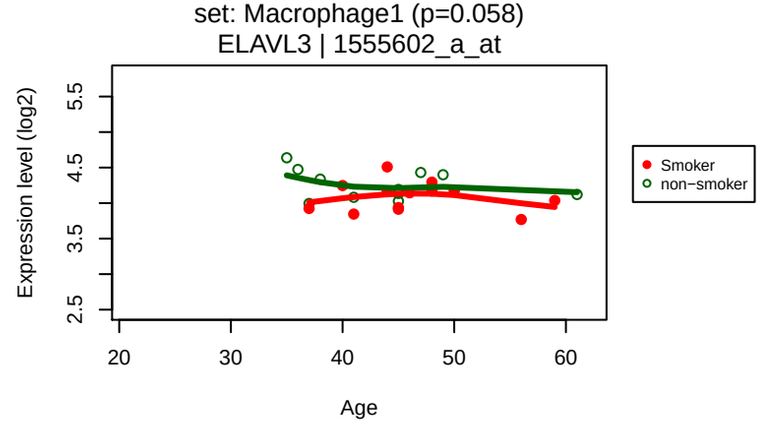

set: SmallAirway1 (p=0.1994)
ELAVL3 | 1555602_a_at
Expression level (log2)
Age
Smoker
non−smoker

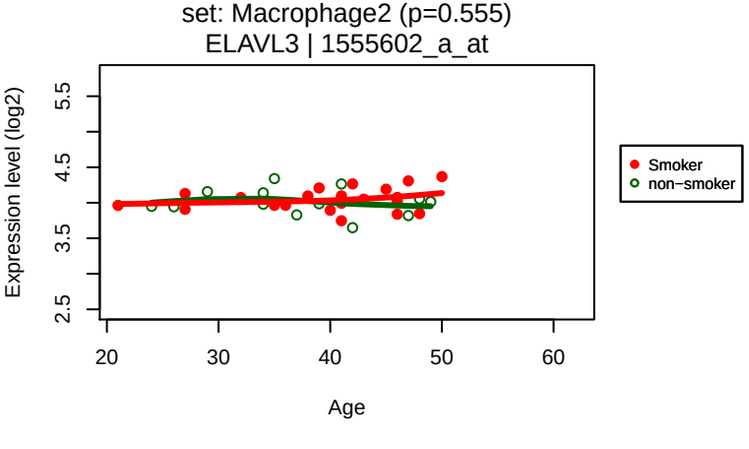

set: SmallAirway2 (p=0.3086)
ELAVL3 | 1555602_a_at
Expression level (log2)
Age
Smoker
non−smoker

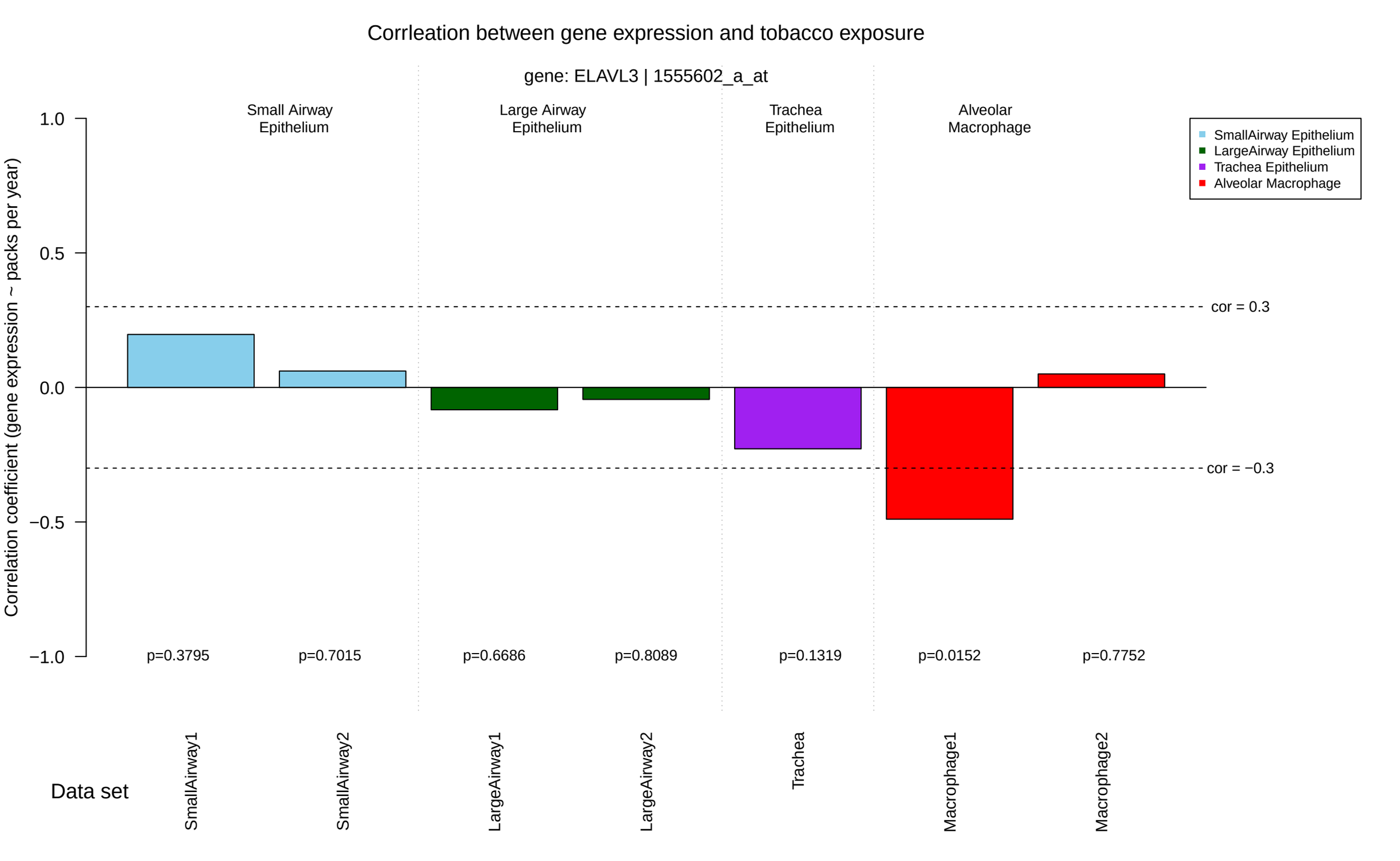

set: LargeAirway1 (p=0.3534)
ELAVL3 | 1555602_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4711)
ELAVL3 | 1555602_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.9711)
ELAVL3 | 1555602_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.058)
ELAVL3 | 1555602_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.555)
ELAVL3 | 1555602_a_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ELAVL3 | 1555602_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3795
p=0.7015
p=0.6686
p=0.8089
p=0.1319
p=0.0152
p=0.7752
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2